

Centroid of Composite shapes

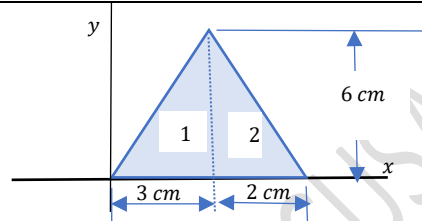
The centroid of a composite shape can be found out by breaking it down to elementary shapes (like triangle, rectangle, circle etc) whose centroids are known. When the area is broken down to small elementary shapes, the centroid is found out by

$$\bar{x} = \frac{\sum x_i \Delta A_i}{\sum \Delta A_i} = \frac{A_1 x_1 + A_2 x_2 + A_3 x_3 + \dots}{A_1 + A_2 + A_3 + \dots}$$

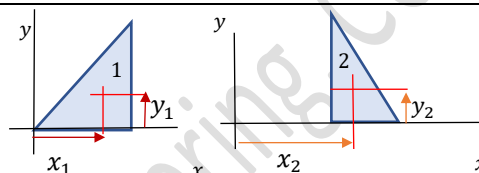
$$\bar{y} = \frac{\sum y_i \Delta A_i}{\sum \Delta A_i} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3 + \dots}{A_1 + A_2 + A_3 + \dots}$$

Example 1

Locate the centroid of the triangle shown with respect to the co-ordinate axes.



The triangle is divided into two right triangles whose centroids are at (x_1, y_1) and (x_2, y_2) with respect to the given x-y axes.



i	Area A_i (cm ²)	x_i (cm)	y_i (cm)	$A_i x_i$	$A_i y_i$
1	$\frac{1}{2} \times 3 \times 6 = 9$	$3 - \frac{1}{3} \times 3 = 2$	$\frac{1}{3} \times 6 = 2$	18	18
2	$\frac{1}{2} \times 2 \times 6 = 6$	$3 + \frac{1}{3} \times 2 = 3.67$	$\frac{1}{3} \times 6 = 2$	22	12
Σ	15			40	30

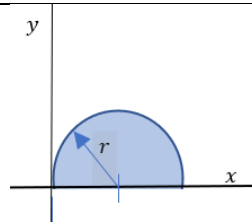
$$\bar{x} = \frac{\sum A_i x_i}{\sum A_i} = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = \frac{40}{15} = 2.67 \text{ cm}$$

$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = \frac{30}{15} = 2 \text{ cm}$$

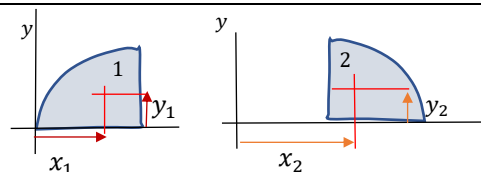
(Notice that $\bar{y}$ of a triangle from the base is always $\frac{h}{3}$ irrespective of it a right triangle or not)

Example 2

Locate the centroid of the semicircle of radius r shown with respect to the co-ordinate axes.



Divide the semi-circle into two quarter circles whose centroids are at (x_1, y_1) and (x_2, y_2) with respect to the given x-y axes



i	Area A_i (cm ²)	x_i (cm)	y_i (cm)	$A_i x_i$	$A_i y_i$
1	$\frac{\pi r^2}{4}$	$r - \frac{4r}{3\pi}$	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{4} \left(r - \frac{4r}{3\pi} \right)$	$\frac{\pi r^2}{4} \left(\frac{4r}{3\pi} \right)$
2	$\frac{\pi r^2}{4}$	$r + \frac{4r}{3\pi}$	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{4} \left(r + \frac{4r}{3\pi} \right)$	$\frac{\pi r^2}{4} \left(\frac{4r}{3\pi} \right)$
Σ	$\frac{\pi r^2}{2}$			$\frac{\pi r^3}{2}$	$\frac{2r^3}{3\pi}$

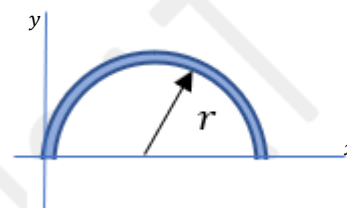
$$\bar{x} = \frac{\Sigma A_i x_i}{\Sigma A_i} = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = \frac{\frac{\pi r^3}{2}}{\frac{\pi r^2}{2}} = r$$

$$\bar{y} = \frac{\Sigma A_i y_i}{\Sigma A_i} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = \frac{\frac{2r^3}{3\pi}}{\frac{\pi r^2}{2}} = \frac{4r}{3\pi}$$

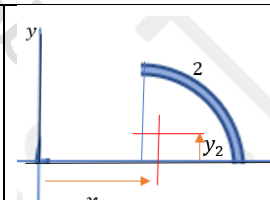
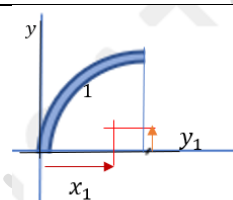
The centroid of the semicircle lies on the symmetrical diameter at a height $\frac{4r}{3\pi}$ from the bounding diameter.

Example 3

Locate the centroid of the semicircular arc of radius r shown with respect to the co-ordinate axes.



Divide the semi-circular arc in to two quarter circular arcs whose centroids are at (x_1, y_1) and (x_2, y_2) with respect to the given x-y axes.



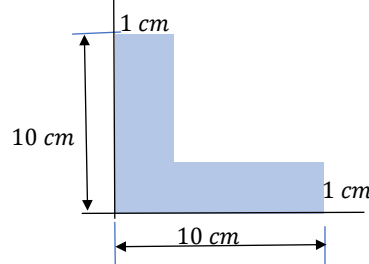
i	Length L_i (cm)	x_i (cm)	y_i (cm)	$L_i x_i$	$L_i y_i$
1	$\frac{2\pi r}{4}$	$r - \frac{2r}{\pi}$	$\frac{2r}{\pi}$	$\frac{\pi r}{2} \left(r - \frac{2r}{\pi} \right)$	$\frac{\pi r}{2} \left(\frac{2r}{\pi} \right)$
2	$\frac{2\pi r}{4}$	$r + \frac{2r}{\pi}$	$\frac{2r}{\pi}$	$\frac{\pi r}{2} \left(r + \frac{2r}{\pi} \right)$	$\frac{\pi r}{2} \left(\frac{2r}{\pi} \right)$
Σ	πr			πr^2	$2r^2$

$$\bar{x} = \frac{\Sigma L_i x_i}{\Sigma L_i} = \frac{L_1 x_1 + L_2 x_2}{L_1 + L_2} = \frac{\pi r^2}{\pi r} = r$$

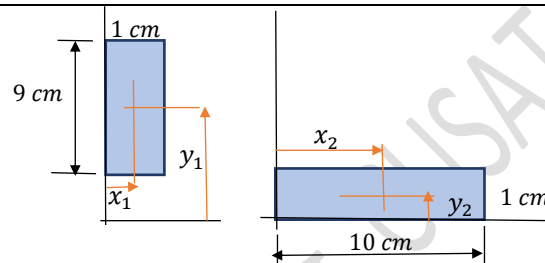
$$\bar{y} = \frac{\Sigma L_i y_i}{\Sigma L_i} = \frac{L_1 y_1 + L_2 y_2}{L_1 + L_2} = \frac{2r^2}{\pi r} = \frac{2r}{\pi}$$

Example 4

Locate the centroid of the angle section shown in the figure with respect to the co-ordinate axes shown.



Divide the given section in to two rectangles with centroids at (x_1, y_1) and (x_2, y_2) with respect to the given x-y axes.



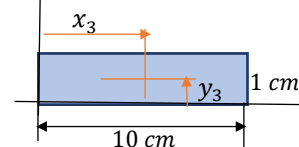
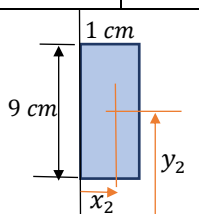
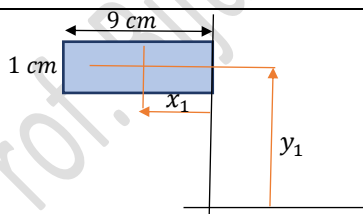
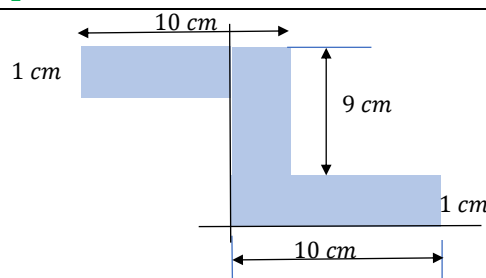
i	Area A_i (cm ²)	x_i (cm)	y_i (cm)	$A_i x_i$	$A_i y_i$
1	$1 \times 9 = 9$	$\frac{1}{2} \times 1 = 0.5$	$1 + \frac{1}{2} \times 9 = 5.5$	4.5	49.5
2	$10 \times 1 = 10$	$\frac{1}{2} \times 10 = 5$	$\frac{1}{2} \times 1 = 0.5$	50	5
Σ	19			54.5	54.5

$$\bar{x} = \frac{\Sigma A_i x_i}{\Sigma A_i} = \frac{A_1 x_1 + A_2 x_2}{A_1 + A_2} = \frac{54.9}{19} = 2.9 \text{ cm}$$

$$\bar{y} = \frac{\Sigma A_i y_i}{\Sigma A_i} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} = \frac{54.9}{19} = 2.9 \text{ cm}$$

Example 5

Locate the centroid of the section shown in the figure with respect to the co-ordinate axes shown.



Divide the given section in to three rectangles with centroids at (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) with respect to the given x-y axes.

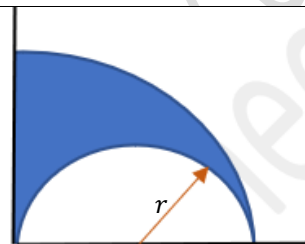
i	Area A_i (cm ²)	x_i (cm)	y_i (cm)	$A_i x_i$	$A_i y_i$
1	$1 \times 9 = 9$	$-\frac{1}{2} \times 9 = -4.5$	$10 + \frac{1}{2} \times 1 = 10.5$	-40.5	94.5
2	$9 \times 1 = 9$	$\frac{1}{2} \times 1 = 0.5$	$1 + \frac{1}{2} \times 9 = 5.5$	4.5	49.5
3	$10 \times 1 = 10$	$\frac{1}{2} \times 10 = 5$	$\frac{1}{2} \times 1 = 0.5$	50	5
Σ	28			14	149

$$\bar{x} = \frac{\Sigma A_i x_i}{\Sigma A_i} = \frac{14}{28} = 0.5 \text{ cm}$$

$$\bar{y} = \frac{\Sigma A_i y_i}{\Sigma A_i} = \frac{149}{28} = 5.3 \text{ cm}$$

Example 6

Locate the centroid of the shaded area in which a semi-circular area with radius r is removed from a quarter circular area of radius $2r$.



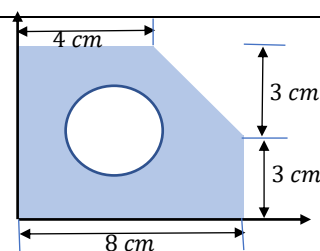
i	Area A_i (cm ²)	x_i (cm)	y_i (cm)	$A_i x_i$	$A_i y_i$
1	$\frac{\pi(2r)^2}{4}$	$\frac{4(2r)}{3\pi}$	$\frac{4(2r)}{3\pi}$	$\frac{4\pi r^2}{4} \left(\frac{8r}{3\pi}\right) = 2.67r^3$	$\frac{4\pi r^2}{4} \left(\frac{8r}{3\pi}\right) = 2.67r^3$
2	$-\frac{\pi r^2}{2}$	r	$\frac{4r}{3\pi}$	$-\frac{\pi r^3}{2} = -1.57r^3$	$-\frac{\pi r^2}{2} \left(\frac{4r}{3\pi}\right) = -0.67r^3$
Σ	$1.57r^2$			$1.1r^3$	$2r^3$

$$\bar{x} = \frac{\Sigma A_i x_i}{\Sigma A_i} = \frac{1.1r^3}{1.57r^2} = 0.7r$$

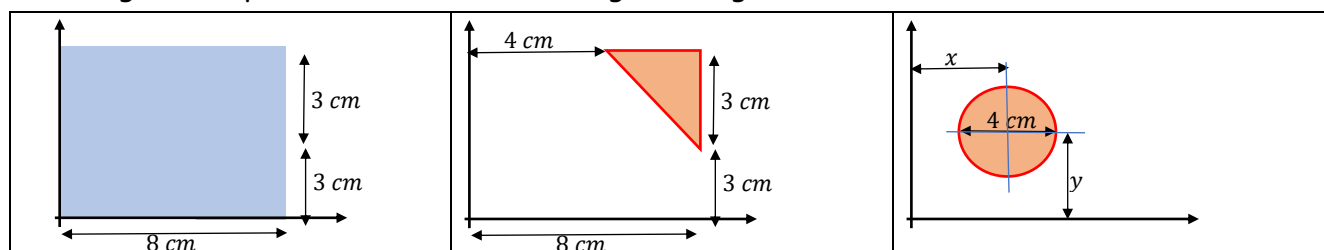
$$\bar{y} = \frac{\Sigma A_i y_i}{\Sigma A_i} = \frac{2r^3}{1.57r^2} = 1.27r$$

Example 7

Find the co-ordinates of the centre of a 4cm diameter circular hole cut in a thin plate so that this point will be the centroid of the remaining shaded area.



Let the given shape be divided into a rectangle, triangle, and circle as shown below.



Let the co-ordinates of the centre of the circle be (x, y) .

i	Area A_i (cm^2)	x_i (cm)	y_i (cm)	$A_i x_i$	$A_i y_i$
1	$8 \times 6 = 48$	4	3	192	144
2	$-\frac{1}{2} \times 4 \times 3 = -6$	$8 - \frac{1}{3} \times 4 = 6.67$	$6 - \frac{1}{3} \times 3 = 5$	-40	-30
3	$-\frac{\pi \times 4^2}{4} = -4\pi$	x	y	$-4\pi x$	$-4\pi y$
Σ	29.43			$152 - 4\pi x$	$114 - 4\pi y$

The centroid of the shaded area is given by

$$\bar{x} = \frac{\Sigma A_i x_i}{\Sigma A_i} = \frac{152 - 4\pi x}{29.43} \quad \ll \gg \quad \bar{y} = \frac{\Sigma A_i y_i}{\Sigma A_i} = \frac{114 - 4\pi y}{29.43}$$

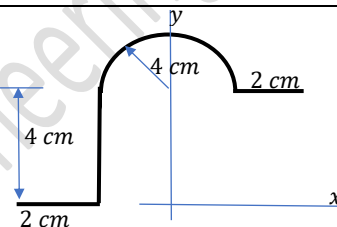
It is given that the centre of the circle removed is the centroid of the remaining shaded area.

$$x = \frac{152 - 4\pi x}{29.43} \gg \gg x = 3.62 \text{ cm}$$

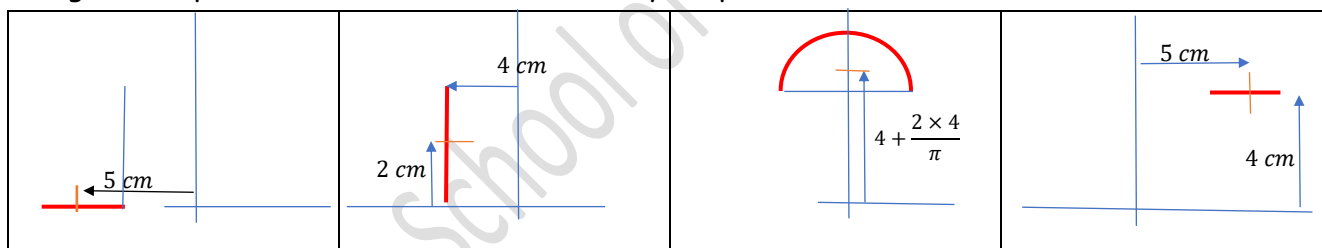
$$y = \frac{114 - 4\pi y}{29.43} \gg \gg y = 2.71 \text{ cm}$$

Example 8

Locate the centroid of the wire bent in the shape as shown.



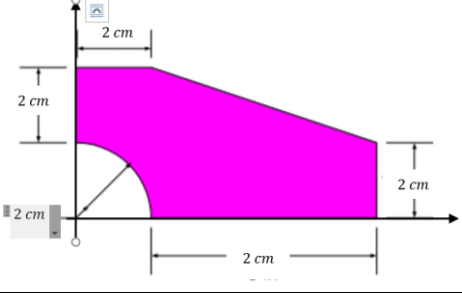
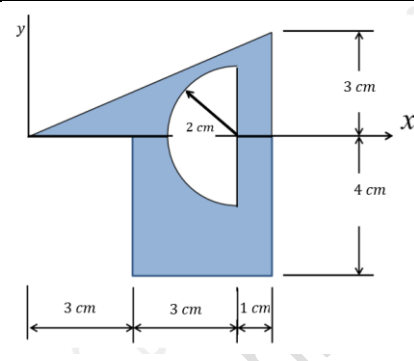
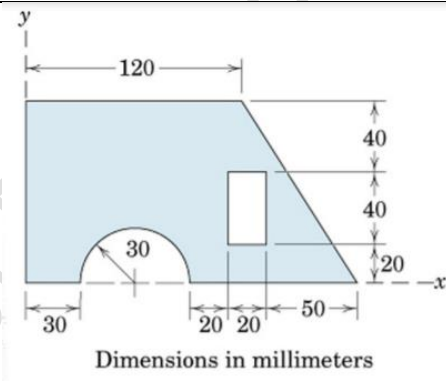
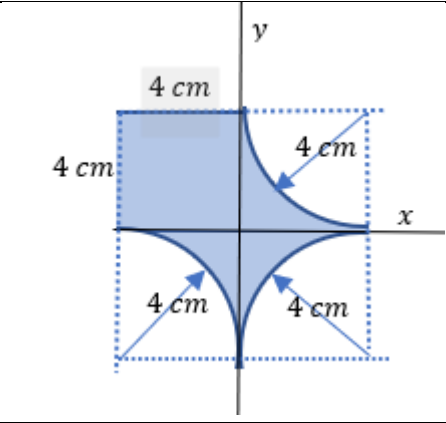
The given shape is divided into four elementary shapes



i	Length L_i (cm)	x_i (cm)	y_i (cm)	$L_i x_i$	$L_i y_i$
1	2	-5	0	-10	0
2	4	-4	2	-16	8
3	$\frac{2\pi \times 4}{2} = 4\pi$	0	$4 + \frac{2 \times 4}{\pi} = 6.55$	0	82.31
4	2	5	4	10	8
Σ	20.57			-16	98.31

$$\bar{x} = \frac{\Sigma A_i x_i}{\Sigma A_i} = \frac{-16}{20.57} = -0.77 \text{ cm} \quad \ll \gg \quad \bar{y} = \frac{\Sigma A_i y_i}{\Sigma A_i} = \frac{98.31}{20.57} = 4.78 \text{ cm}$$

#Exercise 12

12.1 Locate the centroid of the shaded area with respect to the co-ordinate axes shown.	
12.2 Locate the centroid of the shaded area with respect to the co-ordinate axes shown.	
12.3 Locate the centroid of the shaded area with respect to the co-ordinate axes shown.	 Dimensions in millimeters
12.4 Locate the centroid of the shaded area with respect to the co-ordinate axes shown.	

Reference

1. Engineering Mechanics: Timoshenko and Young, McGraw Hill
2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley

4. Engineering Mechanics: Hibbeler, Pearson

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